

16 May 2022

Series Solution

▣ Define Ordinary point and singular point

* Consider differential Equation:

$$P_0(x) \frac{d^2 y}{dx^2} + P_1(x) \frac{dy}{dx} + P_2(x) y = 0 \quad \text{--- (1)}$$

A point $x=a$ is called ordinary point of DE (1) if $P_0(x) \neq 0$ at $x=a$ i.e, $P_0(a) \neq 0$

Example:

$$5 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + 4x^2 y = 0$$

Here, $x=0$ is an ordinary point.

* Consider differential Equation.

$$\cancel{P_0(x) = 0} \quad P_0(x) \frac{d^2 y}{dx^2} + P_1(x) \frac{dy}{dx} + P_2(x) y = 0 \quad \text{--- (1)}$$

A point $x=a$ is called singular point of DE (1)

if $P_0(x) = 0$ at $x=a$ i.e, $P_0(a) = 0$

Example:

$$x^2 \frac{d^2 y}{dx^2} + (2x+1) \frac{dy}{dx} + y = 0$$

Here, $x=0$ is a singular point.

Q. Solve $\frac{d^2y}{dx^2} + xy = 0$ by series method.

Solution: Given differential equation, $\frac{d^2y}{dx^2} + xy = 0$ — (i)

Here, $P_0(x) = 1$

$$\therefore P_0(0) = 1 \neq 0$$

i.e, $x=0$ is an ordinary point of given differential equation.

Consider the series solution of (i) as

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n + \dots \quad (ii)$$

Differentiating (ii) with respect to x

$$\frac{dy}{dx} = a_1 + 2a_2x + 3a_3x^2 + \dots + na_nx^{n-1} + \dots$$

$$\therefore \frac{d^2y}{dx^2} = 2a_2 + 6a_3x + \dots + n(n-1)a_nx^{n-2} + \dots \quad (iii)$$

Putting (ii) and (iii) in equation (i)

$$\begin{aligned} & [2a_2 + 6a_3x + \dots + n(n-1)a_nx^{n-2} + \dots] + x[a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n] = 0 \\ \Rightarrow & [2a_2 + 6a_3x + \dots + n(n-1)a_nx^{n-2} + \dots] + [a_0x + a_1x^2 + a_2x^3 + a_3x^4 + \dots + a_nx^{n+1} + \dots] = 0 \end{aligned}$$

Equating coefficient of x^0

$$2a_2 = 0$$

$$\therefore a_2 = 0$$

Equating coefficient of x

$$6a_3 + a_0 = 0$$

$$\Rightarrow 6a_3 = -a_0$$

$$\Rightarrow a_3 = -\frac{a_0}{6}$$

Now, equating coefficient of x^2

Recurrence formula

$$(n+2)(n+1)a_{n+2} + a_{n-1} = 0$$

$$\Rightarrow (n+2)(n+1)a_{n+2} = -a_{n-1}$$

$$\Rightarrow a_{n+2} = \frac{-a_{n-1}}{(n+2)(n+1)}$$

Putting $n=2$

$$a_4 = \frac{-a_1}{4 \times 3} = \frac{-a_1}{12}$$

Putting $n=3$,

$$a_5 = \frac{-a_2}{20} = 0 \quad [\because a_2 = 0]$$

Putting $n=4$,

$$a_6 = \frac{-a_3}{30} = -\frac{1}{30} \left(-\frac{a_0}{6} \right) = \frac{a_0}{180}$$

Equating coefficient of x

$$6a_3 + a_0 = 0$$

$$\Rightarrow 6a_3 = -a_0$$

$$\therefore a_3 = -\frac{a_0}{6}$$

Putting $a_2, a_3, a_4, a_5, a_6, \dots$ in equation (i)

$$y = a_0 + a_1x + 0 \cdot x^2 - \frac{a_0}{6}x^2 - \frac{a_1}{12}x^3 + 0 \cdot x^4 + \frac{a_0}{180}x^5 + \dots$$

$$= a_0 + a_1x - \frac{a_0}{6}x^2 - \frac{a_1}{12}x^3 + \frac{a_0}{180}x^5 + \dots$$

$$= \left(1 - \frac{x^2}{6} + \frac{x^5}{180} + \dots \right) a_0 + \left(x - \frac{x^3}{12} + \dots \right) a_1$$

which is the required solution.

[Answer]

Q. Solve $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + 2y = 0$ by series solution.

Solution: Given differential equation $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + 2y = 0$ — (i)

Here, $P_0(x) = 1-x^2$

$\Rightarrow P_0(0) = \frac{1}{1} \neq 0$

i.e, $x=0$ is an ordinary point of given differential equation..
 Cond: Consider the series solution of (i) as

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_nx^n + \dots \quad (ii)$$

Differentiating (ii) with respect to x

$$\frac{dy}{dx} = a_1 + 2a_2x + 3a_3x^2 + \dots + na_nx^{n-1} + \dots \quad (iii)$$

$$\therefore \frac{d^2y}{dx^2} = 2a_2 + 6a_3x + n(n-1)a_nx^{n-2} + \dots \quad (iv)$$

Putting (ii), (iii) and (iv) in equation (i)

$$\begin{aligned} (1-x^2)[2a_2 + 6a_3x + n(n-1)a_nx^{n-2} + \dots] - x[a_1 + 2a_2x + 3a_3x^2 + \dots + na_nx^{n-1}] \\ + x[a_0 + a_1x + a_2x^2 + a_3x^3 + a_nx^n + \dots] \\ \Rightarrow [2a_2 + 6a_3x + n(n-1)a_nx^{n-2} + \dots] + [-2a_2x^2 - 6a_3x^3 - n(n-1)a_nx^n + \dots] \\ + [-a_1x - 2a_2x^2 - 3a_3x^3 - \dots - na_nx^n] + [a_0x + a_1x^2 + a_2x^3 + a_3x^4 + a_nx^{n+1} + \dots] \end{aligned}$$

Equating coefficient of x^0

$$a_2 = -2a_0$$

Equating coefficient of x

$$a_3 = -\frac{a_1}{2}$$

Now, equating coefficient of x^n

$$a_{n+2} = \frac{(n-2)a_n}{n+1}$$

Putting $n=2$,

$$a_4 = \frac{(2-2)a_2}{2+1} = 0$$

Putting $n=3$,

$$a_5 = \frac{(3-2)a_3}{3+1} = \frac{a_3}{4}$$

Putting $n=4$,

$$a_6 = \frac{(4-2)a_4}{4+1} = \frac{2a_4}{5}$$

Putting $a_2, a_3, a_4, a_5, a_6, \dots$ in equation (ii)

$$y = a_0 + a_1x - 2a_0x^2 - \frac{a_1}{2}x^3 + 0 \cdot x^4 + \frac{a_3}{4}x^5 + \frac{2a_4}{5}x^6 + \dots$$

$$= a_0 + a_1x - 2a_0x^2 - \frac{a_1}{2}x^3 + \frac{a_3}{4}x^5 + \frac{2a_4}{5}x^6 + \dots$$

$\neq$
 which is the required solution.

18 May 2023

Type-1 (Moderate)

Ex. $y'' - xy' - y = 0$ solve by series method $y(0)=1$ and $y'(0)=1$

Solution:

Given, $y'' - xy' - y = 0$ — (i)

Here,

$$P_0(x) = 1$$

$$\therefore P_0(0) = 1 \neq 0$$

Therefore, $x=0$ is an ordinary point.

Consider series solution

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n + \dots \quad \text{--- (ii)}$$

$$y' = a_1 + 2a_2x + 3a_3x^2 + \dots + na_nx^{n-1} + \dots \quad \text{--- (iii)}$$

$$y'' = 2a_2 + 6a_3x + \dots + n(n-1)a_nx^{n-2} + \dots \quad \text{--- (iv)}$$

Putting y , y' and y'' in equation (i)

$$[2a_2 + 6a_3x + \dots + n(n-1)a_nx^{n-2} + \dots] - [a_1x + 2a_2x^2 + 3a_3x^3 + \dots + na_nx^n] - [a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n + \dots] = 0$$

Equating coefficient of x^0

$$2a_2 - a_0 = 0$$

$$\Rightarrow a_2 = \frac{a_0}{2}$$

Equating coefficient of x

$$6a_3 - a_1 - a_1 = 0$$

$$\Rightarrow 6a_3 = 2a_1$$

$$\therefore a_3 = \frac{a_1}{3}$$

Equating coefficient of x^n

$$(n+2)(n+1)a_{n+2} - na_n - a_n = 0$$

$$\Rightarrow (n+2)(n+1)a_{n+2} = (n+1)a_n$$

$$\therefore a_{n+2} = \frac{a_n}{n+2} \quad \text{--- (v)}$$

Putting $n=0$ in (v)

$$a_{0+2} = \frac{a_0}{0+2} = \frac{a_0}{2}$$

Putting $n=1$ in (v)

$$a_3 = \frac{a_1}{3}$$

Putting $n=2$ in (v)

$$a_4 = \frac{a_2}{4} = \frac{a_0}{2 \times 4} = \frac{a_0}{8}$$

Putting $n=3$ in (v)

$$a_5 = \frac{a_3}{5} = \frac{a_1}{15}$$

Putting $n=4$ in (v)

$$a_6 = \frac{a_4}{6} = \frac{a_0}{48}$$

Putting a_2, a_3, a_4, a_5, a_6 in equation (ii)

$$y = a_0 + a_1x + \frac{a_0}{2}x^2 + \frac{a_1}{3}x^3 + \frac{a_0}{8}x^4 + \frac{a_1}{15}x^5 + \frac{a_0}{48}x^6 + \dots$$

$$= a_0 \left(1 + \frac{x^2}{2} + \frac{x^4}{8} + \frac{x^6}{48} + \dots \right) + a_1 \left(x + \frac{x^3}{3} + \frac{x^5}{15} + \dots \right) \quad \text{--- (v)}$$

which is the required solution.

Now initial condition $y(0)=1$ and $y'(0)=1$

$$\text{Hence, } y = a_0 \left(1 + \frac{x^2}{2} + \frac{x^4}{8} + \dots \right) + a_1 \left(x + \frac{x^3}{3} + \frac{x^5}{15} + \dots \right)$$

$$y' = a_0 \left(x + \frac{x^3}{2} + \dots \right) + a_1 \left(1 + x^2 + \frac{x^4}{3} + \dots \right)$$

Therefore,

$$y(0) = a_0(1+0) + a_1 \times 0$$

$$\Rightarrow 1 = a_0$$

$$\text{and } y'(0) = a_0 \times 0 + a_1$$

$$\Rightarrow 1 = a_1$$

Putting the value of a_0 and a_1 in (i), we get,

$$y = \left(1 + \frac{x^2}{2} + \frac{x^4}{8} + \dots\right) + \left(x + \frac{x^3}{3} + \frac{x^5}{15} + \dots\right)$$

$$= \left(1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{8} + \frac{x^5}{15} + \dots\right)$$

which is the method required solution.

Ex $(x^2+1)y'' + xy' + 2xy = 0$ solve by series method $y(0)=1$
and $y'(0)=1$.

Solution: Given $(x^2+1)y'' + xy' + 2xy = 0$ — (i)

$$\text{Here, } P_0(x) = x^2 + 1$$

$$\therefore P_0(0) = 1 \neq 0$$

Therefore, $x=0$ is an ordinary point.

Consider series solution,

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n + \dots \quad \text{--- (ii)}$$

$$y' = a_1 + 2a_2x + 3a_3x^2 + \dots + (n-1)a_nx^{n-1} + \dots \quad \text{--- (iii)}$$

$$y'' = 2a_2 + 6a_3x + \dots + n(n-1)a_nx^{n-2} + \dots \quad \text{--- (iv)}$$

Putting (ii), (iii), (iv) in equation (i)

$$(x^2+1)[2a_2+6a_3x+\dots+n(n-1)a_nx^{n-2}+\dots]+x[a_1+2a_2x+3a_3x^2+\dots+na_nx^{n-1}]$$

$$+2x[a_0+a_1x+a_2x^2+a_3x^3+\dots+anx^n+\dots]=0$$

$$=[2a_2x^2+2a_2+6a_3x^3+6a_3x+\dots+n(n-1)a_nx^n+a_nx^{n-2}]$$

$$+[a_1x+2a_2x^2+3a_3x^3+\dots+na_nx^n]+[2a_0x+2a_1x^2+2a_2x^3+2a_3x^4+\dots+2a_nx^{n+1}]$$

Equating coefficient of x^0

$$2a_2 = 0$$

$$\Rightarrow a_2 = 0$$

Equating coefficient of x

$$6a_3x+a_1+2a_0=0$$

$$\Rightarrow 6a_3x = -a_1-2a_0$$

$$\Rightarrow a_3 = \frac{-(a_1+2a_0)}{6}$$

Equating coefficient of x^n

$$n(n-1)a_n + (n+2)(n+1)a_{n+2} + na_n + 2a_{n-1} = 0$$

$$\Rightarrow (n+2)(n+1)a_{n+2} + a_n\{n(n-1)+n\} + 2a_{n-1} = 0$$

$$\Rightarrow (n+2)(n+1)a_{n+2} = -a_n \cdot n^2 - 2a_{n-1}$$

$$\Rightarrow a_{n+2} = \frac{-(a_n \cdot n^2 + 2a_{n-1})}{(n+2)(n+1)} \quad \text{--- (v)}$$

Putting $n=1$ in equation (v)

$$a_3 = \frac{-(a_1 + 2a_0)}{3 \times 2} = \frac{-(a_1 + 2a_0)}{6}$$

Putting $n=2$ in equation (v)

$$a_4 = \frac{-(a_2 \cdot 4 + 2a_1)}{4 \times 3} = \frac{-(4a_2 + 2a_1)}{12} = \frac{-a_1}{6}$$

Putting $n=3$ in equation (v)

$$a_5 = \frac{-(a_3 \cdot 9 + 2a_2)}{5 \times 4} = \frac{-9\left(\frac{a_1+2a_0}{6}\right) + 2 \cdot 0}{20}$$

$$= \frac{9a_1 + 18a_0}{120} = \frac{9(a_1+2a_0)}{120} = \frac{3a_1+6a_0}{40}$$

Putting a_1, a_2, a_3, a_4, a_5 in equation (ii)

$$y = a_0 + a_1 x + 0 \cdot x^2 + \left(-\frac{a_1+2a_0}{6}\right)x^3 - \frac{a_1}{6}x^4 + \left(\frac{3a_1+6a_0}{40}\right)x^5 + \dots$$

$$= a_0 + a_1 x - \frac{a_1 x^3}{6} - \frac{a_0 x^3}{3} - \frac{a_1}{6}x^4 + \dots \quad \text{--- (vi)}$$

$$\therefore y = a_0 \left(1 - \frac{x^3}{3} + \dots\right) + a_1 \left(x - \frac{x^3}{6} + \dots\right) \quad \text{--- (vii)}$$

which is the required solution.

Now initial condition $y(0)=1$ and $y'(0)=1$

Here,

$$y = a_0 \left(1 - \frac{x^3}{3} + \dots\right) + a_1 \left(x - \frac{x^3}{6} + \dots\right)$$

$$y' = a_0 \left(a_1 x - \frac{a_1 x^3}{6} - \frac{a_0 x^3}{3} - \frac{a_1 x^4}{6} + \dots \right)$$

$$\therefore y' = a_1 - \frac{3a_1 x^2}{6} - \frac{3a_0 x^2}{3} - \frac{4a_1 x^3}{6} + \dots \quad \text{--- (viii)}$$

Now, putting $y(0)$ in (vi) and $y'(0)$ in (viii)

$$y(0) = a_0 + a_1 x - \frac{a_1 x^3}{6} - \frac{a_0 x^3}{3} - \frac{a_1 x^4}{6} + \dots$$

$$\Rightarrow 1 = a_0$$

$$\text{and } y'(0) = a_1 - \frac{3a_1 x^2}{6} - \frac{4a_0 x^2}{3} + \dots$$

$$\Rightarrow 1 = a_1$$

Now putting value of a_0 and a_1 in (vi) we get

$$y = \left(1 - \frac{x^3}{3} + \dots\right) + 1 \left(x - \frac{x^3}{6} + \dots\right)$$

$$= 1 + x - \frac{x^3}{3} - \frac{x^3}{6} + \dots$$

which is the required solution.

28 May 2022

Q. Solve the following Series in power of $x-1$

$$x^2 y'' + x y' + y = 0$$

Solⁿ Given, $x^2 y'' + x y' + y = 0$ — (i)

Let, $v = x-1$ — (ii)

i.e., $v+1 = x$ — (iii)

from (iii) and (i) we get.

$$(v+1)^2 y'' + (v+1) y' + y = 0$$

$$\Rightarrow (v^2 + 2v+1) y'' + (v+1) y' + y = 0$$
 — (iv)

Here $v=0$ is an ordinary point of (iv)

Consider the series solution,

$$y = a_0 + a_1 v + a_2 v^2 + a_3 v^3 + a_4 v^4 + \dots + a_n v^n + \dots$$

$$\therefore \frac{dy}{dv} = a_1 + 2a_2 v + 3a_3 v^2 + 4a_4 v^3 + \dots + n a_n v^{n-1} + \dots$$

$$\text{and, } y'' = 2a_2 + 6a_3 v + 12a_4 v^2 + \dots + n(n-1) a_n v^{n-2} + \dots$$

Putting the value of y, y', y'' into (iv)

$$\begin{aligned} & (v^2 + 2v+1) [2a_2 + 6a_3 v + 12a_4 v^2 + \dots + n(n-1) a_n v^{n-2} + \dots] + (v+1) [a_1 + 2a_2 v + 3a_3 v^2 + 4a_4 v^3 + \dots + n a_n v^{n-1} + \dots] \\ & + [a_0 + a_1 v + a_2 v^2 + a_3 v^3 + a_4 v^4 + \dots + a_n v^n + \dots] = 0 \\ & = [2a_2 v^2 + 6a_3 v^3 + 12a_4 v^4 + \dots + n(n-1) a_n v^n + \dots] + [2a_2 v + 12a_3 v^2 + 24a_4 v^3 + \dots + 2n(n-1) a_n v^{n-1} + \dots] \\ & + [2a_2 + 6a_3 v + 12a_4 v^2 + \dots + n(n-1) a_n v^{n-2}] \\ & + [a_1 v + 2a_2 v^2 + 3a_3 v^3 + 4a_4 v^4 + \dots + n a_n v^{n-1} + \dots] + [a_1 + 2a_2 v + 3a_3 v^2 + 4a_4 v^3 + \dots + n a_n v^{n-1} + \dots] + [a_0 + a_1 v + a_2 v^2 + a_3 v^3 + a_4 v^4 + \dots + a_n v^n + \dots] = 0 \end{aligned}$$

Equating coefficient of v^0

$$2a_2 + a_1 + a_0 = 0$$

$$\therefore a_2 = -\frac{a_1 + a_0}{2}$$

Equating coefficient of v .

$$4a_2 + 6a_3 + a_1 + 2a_2 + a_1 = 0$$

$$\Rightarrow 6a_3 = -6a_2 - 2a_1$$

$$\Rightarrow 3a_3 = -3 \cdot \left(-\frac{a_1 + a_0}{2}\right) - 2a_1$$

$$\Rightarrow 3a_3 = \frac{3a_1 + 3a_0}{2} - 2a_1$$

$$= \frac{3a_1 + 3a_0 - 4a_1}{2}$$

$$\therefore a_3 = \frac{3a_0 + a_1}{6}$$

Recurrence formula

$$n(n-1)a_n + 2(n+1)na_{n+1} + (n+2)(n+1)a_{n+2} + (n+1)a_{n+1} + a_n + na_n = 0$$

$$\Rightarrow a_{n+2} \cdot (n+2)(n+1) = -n(n-1)a_n - a_n - na_n - 2(n+1)na_{n+1} - (n+1)a_{n+1}$$

$$\Rightarrow a_{n+2} (n+2)(n+1) = (-n^2 - n - 1)a_n + (-2n^2 - 2n - n - 1)a_{n+1}$$

$$\therefore a_{n+2} = \frac{(-n^2 - 1)a_n + (-2n^2 - 3n - 1)a_{n+1}}{(n+1)(n+2)} \quad \text{--- (v)}$$

Putting $n=2$ in (v)

$$a_4 = \frac{-5a_2 - 15a_3}{12}$$

$$= -\frac{5}{12} \left(-\frac{a_0 + a_1}{2}\right) - \frac{15}{12} \left(\frac{3a_0 + a_1}{6}\right)$$

$$= -\frac{5}{24} (a_0 + a_1) - \frac{5}{24} (3a_0 + a_1)$$

$$= -\frac{5}{24} (a_0 + a_1 + 3a_0 + a_1)$$

$$= -\frac{5}{24} (-2a_1)$$

$$= -\frac{5}{12} a_0$$

Therefore the series solution is

$$\begin{aligned}
 y &= a_0 + a_1 v + \left(-\frac{a_1 + a_0}{2}\right) v^2 + \left(\frac{a_1 + 3a_0}{6}\right) v^3 + \left(-\frac{5}{12} a_0\right) v^4 + \dots \\
 &= a_0 \left(1 - \frac{v^2}{2} + \frac{v^3}{2} + \frac{5v^4}{12} + \dots\right) + a_1 \left(v + \frac{v^2}{2} + \frac{v^3}{6} + \dots\right) \\
 &= a_0 \left\{1 - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{2} + \frac{5(x-1)^4}{12} + \dots\right\} + a_1 \left\{(x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{6} + \dots\right\}
 \end{aligned}$$

Ex Solve the following series in power of $(x-2)$

$$y'' + (x-1)y' + y = 0$$

30 May 2022

For singular point:

Case-1

if $m_1 \neq m_2$ and $m_1 - m_2 \neq \text{integer}$
then complete solⁿ is

$$y = C_1(y)_{m=m_1} + C_2(y)_{m=m_2}$$

Case-2

if $m_1 = m_2$ the complete solⁿ is

$$y = C_1(y)_{m=m_1} + C_2\left(\frac{\partial y}{\partial m}\right)_{m=m_2}$$

Case-3

if $m_1 \neq m_2$ but $m_1 - m_2 = \text{integer}$
then $y = C_1(y)_{m=m_1} + C_2\left(\frac{\partial y}{\partial m}\right)_{m=m_2}$

Ex Solve $2x^2 \frac{dy}{dx} + x \frac{dy}{dx} - (x+1)y = 0$

Soln: Given, $2x^2 \frac{dy}{dx} + x \frac{dy}{dx} - (x+1)y = 0$ — (i)

Here, $P_0(x) = 2x^2$

at $x=0$, $P_0(x) = 0$

i.e., $x=0$ is a singular point,

Consider series solution,

$$y = a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + a_3 x^{m+3} + \dots \quad \text{--- (ii)}$$

$$\therefore \frac{dy}{dx} = a_0 m x^{m-1} + a_1 (m+1) x^m + a_2 (m+2) x^{m+1} + a_3 (m+3) x^{m+2} + \dots \quad \text{--- (iii)}$$

and,

$$\frac{d^2y}{dx^2} = a_0 m(m-1)x^{m-2} + a_1(m+1)mx^{m-1} + a_2(m+2)(m+1)x^m + a_3(m+3)(m+2)x^{m+1} + \dots \quad (iv)$$

Putting $y, \frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ into (i)

$$2x^2 [a_0 m(m-1)x^{m-2} + a_1(m+1)mx^{m-1} + a_2(m+2)(m+1)x^m + a_3(m+3)(m+2)x^{m+1} + \dots] \\ + x [a_0 mx^{m-1} + a_1(m+1)x^m + a_2(m+2)x^{m+1} + a_3(m+3)x^{m+2} + \dots] \\ - (x+1) [a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + a_3 x^{m+3} + \dots] = 0$$

$$\Rightarrow [2a_0 m(m-1)x^m + 2a_1(m+1)mx^{m+1} + 2a_2(m+2)(m+1)x^{m+2} + \\ 2a_3(m+3)(m+2)x^{m+3} + \dots] + [a_0 mx^m + a_1(m+1)x^{m+1} + a_2(m+2)x^{m+2} + a_3(m+3)x^{m+3} + \dots] \\ - [a_0 mx^{m+1} + a_1 x^{m+2} + a_2 x^{m+3} + a_3 x^{m+4} + \dots] - [a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + a_3 x^{m+3} + \dots] = 0$$

Equating coefficient of x^m both sides

$$2a_3 m(m-1) + a_0 m - a_0 = 0$$

$$\Rightarrow a_3 (2m^2 - 2m + m - 1) = 0$$

$$\Rightarrow 2m^2 - 2m + m - 1 = 0 \quad [\because a_3 \neq 0]$$

$$\Rightarrow 2m(m-1) + 1(m-1) = 0$$

$$\Rightarrow (m-1)(2m+1) = 0$$

$$\therefore m = 1, -\frac{1}{2}$$

Since, $1 \neq -\frac{1}{2}$ and $1 - (-\frac{1}{2}) = \frac{3}{2}$ which is not an integer.

Therefore complete solution of (v) is

$$y = C_1(y)_{m=1} + C_2(y)_{m=-\frac{1}{2}} \quad (v)$$

Equating coefficient of x^{m+1} from both sides.

$$2a_1(m+1)m + a_1(m+1) - a_0m - a_1 = 0$$

$$\Rightarrow a_1(2m^2 + 2m + m + 1 - 1) = a_0m$$

$$\Rightarrow a_1(2m^2 + 3m) = a_0m$$

$$\therefore a_1 = \frac{a_0m}{2m^2 + 3m}$$

Equating coefficient of x^{m+2} from both side

$$2(m+2)(m+1)a_2 + (m+2)a_2 - a_1 - a_2 = 0$$

$$\Rightarrow a_2(2m^2 + 6m + 4 + m + 2 - 1) = a_1$$

$$\Rightarrow a_2(2m^2 + 7m + 5) = \frac{a_0m}{2m^2 + 3m}$$

$$\therefore a_2 = \frac{a_0m}{(2m^2 + 7m + 5)(2m^2 + 3m)} \text{ etc.}$$

Putting a_1, a_2 into (i)

$$y = a_0x^m + \frac{a_0}{(2m^2 + 3m)}x^{m+1} + \frac{a_0m}{(2m^2 + 7m + 5)(2m^2 + 3m)}x^{m+2} + \dots$$

$$\text{Now, } (y)_{m=1} = a_0x + \frac{a_0}{5}x^2 + \frac{a_0}{70}x^3 + \dots$$

$$\text{and } (y)_{m=-\frac{1}{2}} = a_0x^{-\frac{1}{2}} - a_0x^{\frac{1}{2}} + 2a_0\frac{1}{4}a_1x^{\frac{3}{2}} + \dots$$

Putting these values into (v)

$$y = C_1 \left[a_0x + \frac{1}{5}a_0x^2 + \frac{1}{70}a_0x^3 + \dots \right] + C_2 \left[a_0x^{-\frac{1}{2}} - a_0x^{\frac{1}{2}} + \frac{1}{4}a_0x^{\frac{3}{2}} + \dots \right]$$

which is the required solution

Answer

Solving Series Solution

Step-1 $x=0$ is a singular point.

Step-2 $y = a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + \dots$ Letting solution.

Step-3 $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ finding and putting into given equation.

Step-4 Find m by equating coefficient of x^m .

Step-5 with respect to m , we let complete solution from
Case:

Step-6 Equating coefficient of x^{m+1} and x^{m+2} find a_1 and a_2 in the form of a_0 .

Step-7 Putting a_1 and a_2 into step-2.

Step-8 Finding $(y)_{m=m_1}$ and $(y)_{m=m_2}$.

Step-9 Putting step-8 into step-5.

this did work to find the solution.

$$0 = (L-m)(L-m)$$

$$0 = \{ (L-m) - (L-m) \} (L-m)$$

$$[0 \neq 0] \quad 0 = (L-m) - (L-m)$$

$$0 = (L-m)(L-m)$$

$$L-m = 0$$

$$L = m \quad L-m = 0$$

1 June 2022

▣ Solve $x^2 y'' - xy' + (x^2 + 1)y = 0$ in series method.

Solⁿ Here $x^2 y'' - xy' + (x^2 + 1)y = 0$ ——— (i)

$$\therefore P_0(x) = x^2$$

$$\text{at } x=0, P_0(x) = 0$$

Thus $x=0$ is a regular singular point

Consider series solution

$$y = a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + a_3 x^{m+3} + \dots \quad \text{--- (ii)}$$

$$\therefore y' = a_0 m x^{m-1} + a_1 (m+1) x^m + a_2 (m+2) x^{m+1} + \dots \quad \text{--- (iii)}$$

$$\therefore y'' = a_0 m(m-1) x^{m-2} + a_1 (m+1)m x^{m-1} + a_2 (m+2)(m+1) x^m + \dots \quad \text{--- (iv)}$$

Putting y, y', y'' in (i):

$$\begin{aligned} & x^2 [a_0 m(m-1) x^{m-2} + a_1 (m+1)m x^{m-1} + a_2 (m+2)(m+1) x^m + \dots] \\ & - x [a_0 m x^{m-1} + a_1 (m+1) x^m + a_2 (m+2) x^{m+1} + \dots] \\ & + (x^2 + 1) [a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + \dots] = 0 \\ & = [a_0 m(m-1) x^m + a_1 (m+1)m x^{m+1} + a_2 (m+2)(m+1) x^{m+2} + \dots] \\ & - [a_0 m x^m + a_1 (m+1) x^{m+1} + a_2 (m+2) x^{m+2} + \dots] \\ & + [a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + \dots] + [a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + \dots] = 0 \end{aligned}$$

Equating coefficient of x^m from both side.

$$a_0 m(m-1) - a_0 m + a_0 = 0$$

$$\Rightarrow a_0 \{m(m-1) - m + 1\} = 0$$

$$\Rightarrow m(m-1) - (m-1) = 0 \quad [\because a_0 \neq 0]$$

$$\Rightarrow (m-1)(m-1) = 0$$

$$\therefore m = 1, 1$$

$$\text{i.e., } m_1 = 1, m_2 = 1$$

Hence, $m_1 = m_2$ therefore complete solution is

$$y = C_1 (y)_{m=m_1} + C_2 \left(\frac{\partial y}{\partial m} \right)_{m=m_2} \quad \text{--- (v)}$$

Equating coefficient of x^{m+1} from both sides.

$$a_1(m+1)m - a_1(m+1) + a_1 = 0$$

$$\Rightarrow a_1 \{ m^2 + m - m - 1 + 1 \} = 0$$

$$\Rightarrow a_1 m^2 = 0$$

$$\therefore a_1 = 0 \quad \text{since } m \neq 0$$

Equating coefficient of x^{m+2} from both sides

$$a_2(m+1)(m+2) - a_2(m+2) + a_0 + a_2 = 0$$

$$\Rightarrow a_2 (m^2 + 3m + 2 - m - 2 + 1) = -a_0$$

$$\Rightarrow a_2 (m^2 + 2m + 1) = -a_0$$

$$\Rightarrow a_2 = \frac{-a_0}{(m+1)^2} \text{ etc.}$$

Putting a_1, a_2 in equation (ii)

$$y = a_0 x^m - \frac{a_0}{(m+1)^2} x^{m+2} - \dots$$

$$\text{And, } \frac{\partial y}{\partial m} = a_0 m x^{m-1} + \frac{2a_0}{(m+1)^3} x^{m+2} - \frac{a_0(m+2)}{(m+1)^2} x^{m+1} + \dots$$

(Now, $(y)_{m=1} =$

and $\left(\frac{\partial y}{\partial m} \right)_{m=1} =$

15 June 2022

Solve the follow...

$$\frac{d^2y}{dx^2} + (x-1)\frac{dy}{dx} + y = 0 \text{ in power of } (x-2).$$

Laplace transform:

Let $F(t)$ be a function of t defined for all $t > 0$. then the Laplace transform of $F(t)$ denoted by $\mathcal{L}\{F(t)\}$ or $f(s)$ is defined by

$$\mathcal{L}\{F(t)\} = f(s) = \int_0^\infty e^{-st} F(t) dt$$

provided that the integral exists. s is a parameter which may be real or complex number.

Find Laplace transform, $F(t) = 1$

Solution: We know, from definition of L.T

$$\mathcal{L}\{F(t)\} = \int_0^\infty e^{-st} F(t) dt$$

$$= \int_0^\infty e^{-st} dt$$

$$= \left[\frac{e^{-st}}{-s} \right]_0^\infty$$

$$= \frac{1}{-s} [e^{-st}]_0^\infty$$

$$= \frac{1}{-s} [e^{-\infty} - e^0]$$

$$= \frac{1}{s}$$

Find Laplace transform $F(t)=t$

Solution: We know, from definition of L.T.

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \int_0^{\infty} e^{-st} F(t) dt \\ &= \int_0^{\infty} e^{-st} \cdot t dt \\ &= t \int_0^{\infty} e^{-st} dt - \int_0^{\infty} \left\{ \frac{d}{dt}(t) \right\} \int_0^{\infty} e^{-st} dt \\ &= \left[\frac{t e^{-st}}{-s} \right]_0^{\infty} - \int_0^{\infty} \frac{e^{-st}}{-s} dt \\ &= \frac{1}{s} (0-0) + \frac{1}{s} \left[\frac{e^{-st}}{-s} \right]_0^{\infty} \\ &= -\frac{1}{s} \cdot \frac{1}{s} [e^{-\infty} - e^0] \\ &= \frac{1}{s^2} \end{aligned}$$

Note $e^{-\infty}=0$
 $e^0=1$

Find Laplace transform.

$$\begin{aligned} F(t) &= t^n, \quad F(t) = \sin at, \quad f(t) = \cos at, \\ F(t) &= e^t, \quad F(t) = e^{at} \end{aligned}$$

$F(t) = t^n$

We know, from definition of Laplace transform,

$$\mathcal{L}\{F(t)\} = \int_0^{\infty} e^{-st} F(t) dt$$

$$= \int_0^{\infty} e^{-st} t^n dt$$

$$= t^n \int_0^{\infty} e^{-st} dt - \int_0^{\infty} \left\{ \frac{d}{dt} (t^n) \int_0^{\infty} e^{-st} dt \right\} dt$$

$$= \left[\frac{t^n e^{-st}}{-s} \right]_0^{\infty} - \int_0^{\infty} n t^{n-1} \cdot \frac{e^{-st}}{-s} dt$$

$$= \left[\frac{t^n e^{-st}}{-s} \right]_0^{\infty} - \frac{n}{s} \int_0^{\infty} t^{n-1} e^{-st} dt$$

$$= \left[\frac{t^n e^{-st}}{-s} \right]_0^{\infty} - \frac{n}{s} \cdot \frac{1}{s} \int_0^{\infty} t^{n-1} e^{-st} dt$$

0 = 0 - 0 = 0

and Laplace transform

$$F(t) = t^n \Rightarrow \mathcal{L}\{F(t)\} = \frac{n!}{s^{n+1}}$$

$$F(t) = t^n \Rightarrow \mathcal{L}\{F(t)\} = \frac{n!}{s^{n+1}}$$

25 June, 2022

Some important properties of Laplace transform:

Q1 Linearity property:

If λ_1 and λ_2 are any constants while $f_1(t)$ and $f_2(t)$ are functions with Laplace transform $f_1(s)$ and $f_2(s)$ respectively then,

$$\mathcal{L}\{\lambda_1 f_1(t) + \lambda_2 f_2(t)\} = \lambda_1 \mathcal{L}\{f_1(t)\} + \lambda_2 \mathcal{L}\{f_2(t)\}$$

or, $\lambda_1 f_1(s) + \lambda_2 f_2(s)$

Proof: From the definition of Laplace transform,

$$\mathcal{L}\{f_1(t)\} = f_1(s) = \int_0^{\infty} e^{-st} f_1(t) dt$$

$$\text{and } \mathcal{L}\{f_2(t)\} = f_2(s) = \int_0^{\infty} e^{-st} f_2(t) dt$$

Now for any constants λ_1 and λ_2

$$\begin{aligned}\mathcal{L}\{\lambda_1 f_1(t) + \lambda_2 f_2(t)\} &= \int_0^{\infty} e^{-st} \{\lambda_1 f_1(t) + \lambda_2 f_2(t)\} dt \\ &= \int_0^{\infty} e^{-st} \lambda_1 f_1(t) dt + \int_0^{\infty} e^{-st} \lambda_2 f_2(t) dt \\ &= \lambda_1 \int_0^{\infty} e^{-st} f_1(t) dt + \lambda_2 \int_0^{\infty} e^{-st} f_2(t) dt \\ &= \lambda_1 \mathcal{L}\{f_1(t)\} + \lambda_2 \mathcal{L}\{f_2(t)\} \checkmark \\ &= \lambda_1 f_1(s) + \lambda_2 f_2(s) \checkmark\end{aligned}$$

Hence proved.

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Some important properties of Laplace transform

$$* \mathcal{L}(e^{at}) = \frac{1}{s-a}; [s > a]$$

$$* \mathcal{L}(t^n) = \frac{n!}{s^{n+1}}; [s > 0]$$

$$* \mathcal{L}(\sin at) = \frac{a}{s^2 + a^2}; [s > 0]$$

$$* \mathcal{L}(\cos at) = \frac{s}{s^2 + a^2}$$

Proof: From the definition of Laplace transform

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$$

$$\mathcal{L}\{e^{at}\} = \int_0^\infty e^{-st} e^{at} dt$$

$$= \int_0^\infty e^{-(s-a)t} dt$$

$$= \left[-\frac{e^{-(s-a)t}}{s-a} \right]_0^\infty$$

$$= \frac{1}{s-a} \left[1 - 0 \right]$$

$$= \frac{1}{s-a}$$

$$\mathcal{L}\{t^n\} = \int_0^\infty e^{-st} t^n dt$$

$$= \int_0^\infty e^{-st} t^n dt$$

Let us prove -

Ex Find the laplace transform of $e^{4t} + 4t^3 - 2\sin 3t + 3\cos 5t$

Soln: Applying Linearity property, we have

$$\begin{aligned}\mathcal{L}(e^{4t} + 4t^3 - 2\sin 3t + 3\cos 5t) &= \mathcal{L}(e^{4t}) + 4\mathcal{L}(t^3) - 2\mathcal{L}(\sin 3t) + 3\mathcal{L}(\cos 5t) \\ &= \frac{1}{s-4} + 4 \frac{3!}{s^{3+1}} - 2 \frac{3}{s^2+3^2} + 3 \frac{s}{s^2+5^2} \\ &= \frac{1}{s-4} + \frac{24}{s^4} - \frac{6}{s^2+9} + \frac{3s}{s^2+25}\end{aligned}$$

Ex Find the laplace transform of $3t^4 - 2t^3 + 4e^{-3t} - 2\sin 5t + 3\cos 2t$

Soln: Applying Linearity property, we have.

$$\begin{aligned}\mathcal{L}(3t^4 - 2t^3 + 4e^{-3t} - 2\sin 5t + 3\cos 2t) &= 3\mathcal{L}(t^4) - 2\mathcal{L}(t^3) + 4\mathcal{L}(e^{-3t}) \\ &\quad - 2\mathcal{L}(\sin 5t) + 3\mathcal{L}(\cos 2t) \\ &= 3 \frac{4!}{s^{4+1}} - 2 \frac{3!}{s^{3+1}} + 4 \cdot \frac{1}{s+3} - 2 \cdot \frac{5}{s^2+5^2} + 3 \frac{s}{s^2+2^2} \\ &= \frac{72}{s^5} - \frac{12}{s^4} + \frac{4}{s+3} - \frac{10}{s^2+25} + \frac{3s}{s^2+4}\end{aligned}$$

[Answer]

Ex First shifting or translation property:

$$\text{If } \mathcal{L}\{F(t)\} = f(s) \text{ then } \mathcal{L}\{e^{at}F(t)\} = f(s-a)$$

Proof: By the definition of Laplace transform we have,

$$\mathcal{L}\{F(t)\} = \int_0^{\infty} e^{-st} F(t) dt = f(s)$$

$$\begin{aligned}\therefore \mathcal{L}\{e^{at}F(t)\} &= \int_0^{\infty} e^{-st} e^{at} F(t) dt \\ &= \int_0^{\infty} e^{-(s-a)t} F(t) dt \\ &= f(s-a)\end{aligned}$$

Ex Find the Laplace transform of $t^3 e^{5t}$

Soln:

$$\text{Let, } F(t) = t^3$$

$$\therefore \mathcal{L}\{F(t)\} = \frac{3!}{s^{3+1}}$$

$$\therefore f(s) = \frac{6}{s^4}$$

Thus by 1st shifting property of L.T. we get,

$$\begin{aligned}\mathcal{L}\{e^{5t}t^3\} &= f(s-5) \\ &= \frac{6}{(s-5)^4}\end{aligned}$$

[Answer]

* Find the L.T. of $e^{3t} \{2\cos 5t - 3\sin 5t\}$

Soln Let $f(t) = 2\cos 5t - 3\sin 5t$

$$\therefore \mathcal{L}\{f(t)\} = 2 \cdot \frac{s}{s^2+5^2} - 3 \cdot \frac{5}{s^2+5^2}$$

$$\therefore f(s) = \frac{2s}{s^2+25} - \frac{15}{s^2+25}$$

Thus by 1st shifting property of L.T. we get

$$\mathcal{L}\{e^{3t} \{2\cos 5t - 3\sin 5t\}\} = f(s-3) \\ = \frac{2(s-3) - 15}{(s-3)^2 + 25}$$

$$= \frac{2s - 6 - 15}{s^2 - 6s + 9 + 25}$$

$$= \frac{2s - 21}{s^2 - 6s + 34}$$

[Answer]

30 July, 2022

Partial Differential Equation:

$$p = \frac{\partial z}{\partial x}$$

$$q = \frac{\partial z}{\partial y}$$

$$r = \frac{\partial^2 z}{\partial x^2}$$

$$s = \frac{\partial^2 z}{\partial x \partial y}$$

$$t = \frac{\partial^2 z}{\partial y^2}$$

Charpit Auxiliary Equations:

$$\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{dF}{0}$$

Solve $z - px - qy = p^2 + q^2$ by charpit method

Soln Given, $z - px - qy = p^2 + q^2$

$$\text{Thus } f \equiv z - px - qy - p^2 - q^2 = 0$$

$$\frac{\partial f}{\partial x} = -p$$

$$\frac{\partial f}{\partial y} = -q$$

$$\frac{\partial f}{\partial z} = 1$$

$$\frac{\partial f}{\partial p} = -x - 2p$$

$$\frac{\partial f}{\partial q} = -y - 2q$$

From Charpit's auxiliary equations.

$$\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dz}{-\frac{\partial f}{\partial p}} = \frac{\frac{dy}{y}}{-\frac{\partial f}{\partial q}} = \frac{dF}{0}$$

$$\Rightarrow \frac{dp}{-p+p} = \frac{dq}{-q+q} = \frac{dz}{xp+2p^2+qy+2q^2} = \frac{dz}{x+2p} = \frac{\frac{dy}{y}}{y+2q} = \frac{dF}{0}$$

$$\Rightarrow \frac{dp}{0} = \frac{dq}{0} = \frac{dz}{xp+2p^2+2qy+2q^2} = \frac{dz}{x+2p} = \frac{\frac{dy}{y}}{y+2q} = \frac{dF}{0}$$

From, 1st eqn $\frac{dp}{dp=0}$

Integrating, $p=a$ where a is an arbitrary constant.

From, 2nd eqn $\frac{dq}{dq=0}$

Integrating, $q=b$ where b is an arbitrary constant.

From, $dz = p dx + q dy$ we get

$$dz = a dx + b dy$$

Integrating we get,

$$\int dz = \int a dx + \int b dy$$

$$\Rightarrow z = ax + by + c \text{ where } c \text{ is an arbitrary constant.}$$

which is the complete integral of given equation.

Ex Solve $z = px + qy + pq$ by Charpit method. mod

Given, $z = px + qy + pq$

Thus $f \equiv px + qy + pq - z = 0$

$$\frac{\partial f}{\partial x} = p, \quad \frac{\partial f}{\partial y} = q, \quad \frac{\partial f}{\partial z} = -1, \quad \frac{\partial f}{\partial p} = x + q, \quad \frac{\partial f}{\partial q} = y + p$$

From Charpit Auxiliary Equations:

$$\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{df}{0}$$

$$\Rightarrow \frac{dp}{p - p} = \frac{dq}{q - q} = \frac{dz}{-px - pq - pq - qy} = \frac{dx}{-x - q} = \frac{dy}{-y - p} = \frac{df}{0}$$

$$\Rightarrow \frac{dp}{0} = \frac{dq}{0} = \frac{dz}{-px - 2pq - qy} = \frac{dx}{-x - q} = \frac{dy}{-y - p} = \frac{df}{0}$$

4 August, 2022

Find out Charpit's auxiliary Equation:

Soln Let the partial differential Equation given by $f(x, y, z, p, q) = 0$

Also we have $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$

$$\text{i.e., } dz = p dx + q dy \quad \text{--- (i)}$$

Let us suppose that a relation $F(x, y, z, p, q) = 0$ --- (ii) exists such that after solving (i) and (ii) simultaneously for p and q and putting these values in (i) becomes integrable.

Thus p, q and z may be expressed as functions of x and y . Differentiating partially (i) and (ii) with respect to x

$$\left(\frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial x} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial x} + \frac{\partial f}{\partial p} \cdot \frac{\partial p}{\partial x} + \frac{\partial f}{\partial q} \cdot \frac{\partial q}{\partial x} = 0 \right)$$

$$\Rightarrow \frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} \cdot p + \frac{\partial f}{\partial p} \cdot \frac{\partial p}{\partial x} + \frac{\partial f}{\partial q} \cdot \frac{\partial q}{\partial x} = 0$$

$$\Rightarrow \frac{\partial f}{\partial p} \cdot \frac{\partial p}{\partial x} = -\frac{\partial f}{\partial x} - p \frac{\partial f}{\partial z} - \frac{\partial f}{\partial q} \cdot \frac{\partial q}{\partial x}$$

$$\Rightarrow \frac{\partial p}{\partial x} = \frac{-\frac{\partial f}{\partial x} - p \frac{\partial f}{\partial z} - \frac{\partial f}{\partial q} \cdot \frac{\partial q}{\partial x}}{\frac{\partial f}{\partial p}} \quad \text{--- (iv)}$$

$$\text{Similarly, } \frac{\partial p}{\partial y} = \frac{-\frac{\partial f}{\partial y} - p \frac{\partial f}{\partial z} - \frac{\partial f}{\partial q} \cdot \frac{\partial z}{\partial y}}{\frac{\partial f}{\partial p}} \quad \text{--- (v)}$$

Again differentiating partially (i) and (ii) with respect to y .

$$\frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial y} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial y} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial y} + \frac{\partial f}{\partial p} \cdot \frac{\partial p}{\partial y} + \frac{\partial f}{\partial q} \cdot \frac{\partial q}{\partial y} = 0$$

$$\Rightarrow \frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z} + \frac{\partial f}{\partial p} \cdot \frac{\partial p}{\partial y} + \frac{\partial f}{\partial q} \cdot \frac{\partial q}{\partial y} = 0$$

$$\Rightarrow \frac{\partial f}{\partial q} \cdot \frac{\partial q}{\partial y} = -\frac{\partial f}{\partial y} - q \frac{\partial f}{\partial z} - \frac{\partial f}{\partial p} \cdot \frac{\partial p}{\partial y}$$

$$\therefore \frac{\partial q}{\partial y} = \frac{-\frac{\partial f}{\partial y} - q \frac{\partial f}{\partial z} - \frac{\partial f}{\partial p} \cdot \frac{\partial p}{\partial y}}{\frac{\partial f}{\partial q}} \quad \text{--- (vi)}$$

$$\text{Similarly, } \frac{\partial q}{\partial y} = \frac{-\frac{\partial F}{\partial y} - q \frac{\partial F}{\partial z} - \frac{\partial F}{\partial p} \cdot \frac{\partial p}{\partial y}}{\frac{\partial F}{\partial q}} \quad \text{--- (vii)}$$

From (vi) and (vii) we get,

$$\frac{-\frac{\partial f}{\partial x} - p \frac{\partial f}{\partial z} - \frac{\partial f}{\partial q} \cdot \frac{\partial q}{\partial x}}{\frac{\partial f}{\partial p}} = \frac{-\frac{\partial F}{\partial x} - p \frac{\partial F}{\partial z} - \frac{\partial F}{\partial q} \cdot \frac{\partial q}{\partial x}}{\frac{\partial F}{\partial p}}$$

$$\Rightarrow \frac{\partial f}{\partial x} \cdot \frac{\partial F}{\partial p} + p \frac{\partial f}{\partial z} \cdot \frac{\partial F}{\partial p} + \frac{\partial f}{\partial q} \cdot \frac{\partial q}{\partial x} \cdot \frac{\partial F}{\partial p} = \frac{\partial F}{\partial x} \cdot \frac{\partial f}{\partial p} + p \frac{\partial F}{\partial z} \cdot \frac{\partial f}{\partial p} + \frac{\partial F}{\partial q} \cdot \frac{\partial q}{\partial x} \cdot \frac{\partial f}{\partial p}$$

$$\Rightarrow \left(\frac{\partial f}{\partial x} \cdot \frac{\partial F}{\partial p} - \frac{\partial F}{\partial x} \cdot \frac{\partial f}{\partial p} \right) + p \left(\frac{\partial f}{\partial z} \cdot \frac{\partial F}{\partial p} - \frac{\partial F}{\partial z} \cdot \frac{\partial f}{\partial p} \right) + \frac{\partial q}{\partial x} \left(\frac{\partial f}{\partial q} \cdot \frac{\partial F}{\partial p} - \frac{\partial F}{\partial q} \cdot \frac{\partial f}{\partial p} \right) = 0 \quad \text{--- (viii)}$$

From (vi) and (vii)

$$\frac{-\frac{\partial f}{\partial y} - q \frac{\partial f}{\partial z} - \frac{\partial f}{\partial p} \cdot \frac{\partial p}{\partial y}}{\frac{\partial f}{\partial q}} = \frac{-\frac{\partial F}{\partial y} - q \frac{\partial F}{\partial z} - \frac{\partial F}{\partial p} \cdot \frac{\partial p}{\partial y}}{\frac{\partial F}{\partial q}}$$

$$\Rightarrow \frac{\partial f}{\partial y} \cdot \frac{\partial F}{\partial q} + q \frac{\partial f}{\partial z} \cdot \frac{\partial F}{\partial q} + \frac{\partial f}{\partial p} \cdot \frac{\partial p}{\partial y} \cdot \frac{\partial F}{\partial q} = \frac{\partial F}{\partial y} \cdot \frac{\partial f}{\partial q} + q \frac{\partial F}{\partial z} \cdot \frac{\partial f}{\partial q} + \frac{\partial F}{\partial p} \cdot \frac{\partial p}{\partial y} \cdot \frac{\partial f}{\partial q}$$

$$\Rightarrow \left(\frac{\partial f}{\partial y} \cdot \frac{\partial F}{\partial q} - \frac{\partial F}{\partial y} \cdot \frac{\partial f}{\partial q} \right) + q \left(\frac{\partial f}{\partial z} \cdot \frac{\partial F}{\partial q} - \frac{\partial F}{\partial z} \cdot \frac{\partial f}{\partial q} \right) + \frac{\partial p}{\partial y} \left(\frac{\partial f}{\partial p} \cdot \frac{\partial F}{\partial q} - \frac{\partial F}{\partial p} \cdot \frac{\partial f}{\partial q} \right) = 0 \quad \text{--- (ix)}$$

Hence adding (viii) and (ix), we get after rearranging the terms

$$\left(\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z} \right) \frac{\partial F}{\partial p} + \left(\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z} \right) \frac{\partial F}{\partial q} + \left(-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q} \right) \frac{\partial F}{\partial z} + \left(-\frac{\partial f}{\partial p} \right) \frac{\partial F}{\partial x} + \left(-\frac{\partial f}{\partial q} \right) \frac{\partial F}{\partial y} = 0$$

This is the linear equation of order one with x, y, z, p, q as independent and F as dependent variable.

Therefore as in Lagrangian Method the auxiliary equation are which is the Charpit's auxiliary equation.

iii Solve $z = px + qy + pq$ by Charpit's Method.

Solve Given, $z = px + qy + pq$

Thus, $f \equiv z - px - qy - pq = 0$

$$\frac{\partial f}{\partial x} = -p \quad ; \quad \frac{\partial f}{\partial y} = -q \quad ; \quad \frac{\partial f}{\partial z} = 1, \quad \frac{\partial f}{\partial p} = -x - q, \quad \frac{\partial f}{\partial q} = -y - p$$

Putting these values in Charpit's auxiliary equation,

$$\Rightarrow \frac{dp}{-p+p} = \frac{dq}{-q+q} = \frac{dz}{px+pq+qy+pq} = \frac{dx}{x+q} = \frac{dy}{y+p} = \frac{dF}{0}$$

$$\Rightarrow \frac{dp}{0} = \frac{dq}{0} = \frac{dz}{px+2pq+qy} = \frac{dx}{x+q} = \frac{dy}{y+p} = \frac{dF}{0}$$

From 1st we get,

$$dp = 0$$

$$\Rightarrow \int dp = 0$$

$\Rightarrow p = a$ where a is an arbitrary constant.

From 2nd we get,

$$dq = 0$$

$$\Rightarrow \int dq = 0$$

$\Rightarrow q = b$ where b is an arbitrary constant

Putting the value of p and q into

$$dz = p dx + q dy$$

$$\Rightarrow dz = a dx + b dy$$

Integrating we get,

$$\int dz = a \int dx + b \int dy$$

$\Rightarrow z = ax + by + c$ where c is an arbitrary constant

13 August, 2022

Find the solve solution of $px + qy = pq$ by charpit's method
Soln Given, $px + qy = pq$ — (i)

$$\therefore f \equiv px + qy - pq = 0 \text{ — (ii)}$$

Now,

$$\frac{\partial f}{\partial x} = p$$

$$\frac{\partial f}{\partial y} = q$$

$$\frac{\partial f}{\partial z} = 0$$

$$\frac{\partial f}{\partial p} = x - q$$

$$\frac{\partial f}{\partial q} = y - p$$

from charpit's ~~max~~ auxiliary Equation.

$$\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{df}{0}$$
$$\Rightarrow \frac{dp}{p} = \frac{dq}{q} = \frac{dz}{-px + pq - qy + pq} = \frac{dx}{q-x} = \frac{dy}{p-y} = \frac{df}{0}$$

From 1st two, $\frac{dp}{p} = \frac{dq}{q}$

Integrating we get,

$$\log p = \log q + \log a \text{ where } a \text{ is an arbitrary constant.}$$

$$\Rightarrow \log p = \log qa$$

$$\therefore p = qa \text{ — (iii)}$$

putting the value of p in (i)

$$q(ax+y) = q^2 a$$

$$\Rightarrow ax+y = q a$$

$$\text{i.e. } q = \frac{ax+y}{a} \quad \text{--- (iv)}$$

again from (iii) and (iv)

$$p = \frac{ax+y}{a} - xa$$

$$= ax+y \quad \text{--- (v)}$$

(Now, putting the value of p and q into $dz = p dx + q dy$)

$$\Rightarrow dz = (ax+y) dx + \left(\frac{ax+y}{a}\right) dy$$

$$\Rightarrow a dz = a(ax+y) dx + (ax+y) dy$$

$$= (ax+y)(a dx + dy)$$

Integrating, we get

$$a z = \frac{(ax+y)^2}{2} + c \quad \text{where } c \text{ is an arbitrary constant.}$$

Let, $ax+y = m$

$$\therefore a dx + dy = dm$$

$$\int (ax+y)(a dx + dy)$$

$$= \int m dm$$

$$= \frac{m^2}{2}$$

$$= \frac{(ax+y)^2}{2}$$

Find the solⁿ of $2xz - px^2 + pq = 2qxy$ by charpit's method.

$$\therefore f \equiv 2xz - px^2 + pq - 2qxy = 0 \quad \text{--- (i)}$$

$$\frac{\partial f}{\partial x} = 2z - 2p - 2qy$$

$$\frac{\partial f}{\partial y} = -2qx$$

$$\frac{\partial f}{\partial z} = 2x$$

$$\frac{\partial f}{\partial p} = -2x + q$$

$$\frac{\partial f}{\partial q} = p - 2xy$$

From charpit's auxiliary equation.

$$\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}}$$

$$= \frac{dp}{2z - 2qy} = \frac{dq}{0} = \frac{dx}{x^2 - q} = \frac{dy}{2xy - p} = \frac{dz}{px^2 + 2xyq - qp} = \frac{df}{0}$$

From second

$$dq = 0$$

Integrating we get, $q = a$ where a is an arbitrary constant

putting $q = a$ in (i)

$$2xz - px^2 + pa = 2axy$$

$$\Rightarrow p(a - x^2) = 2axy - 2xz$$

$$\therefore p = \frac{2x(ay - z)}{a - x^2}$$

putting the value of p and q into

$$dz = p dx + q dy$$

$$\Rightarrow dz = \frac{2x(ay - z)dx}{a - x^2} + a dy$$

$$\Rightarrow \frac{dz}{a - x^2} = \frac{2x dx}{a - x^2} + \frac{a dy}{a - x^2}$$

$$\Rightarrow \frac{dz}{ay - z} = \frac{2x dx}{a - x^2} + \frac{a dy}{ay - z}$$

$$\Rightarrow \frac{dz}{ay - z} - \frac{a dy}{ay - z} = \frac{2x dx}{a - x^2}$$

$$\Rightarrow \frac{dz - a dy}{ay - z} = \frac{2x dx}{a - x^2}$$

Integrating we get,

$$- \log(ay - z) = - \log(a - x^2) + \log b$$

$$\Rightarrow \log(ay - z) = \log(a - x^2) - \log b$$

$$\Rightarrow \log(ay - z) = \log \frac{a - x^2}{b}$$

$$\therefore ay - z = \frac{a - x^2}{b}$$

Find the solution $p(1 + q^2) = q(z - a)$ by Charpit's method.

Given that,

$$P(1 + q^2) = q(z - a)$$

8 September, 2022

* Solve $z - px - qy = p^2 + q^2$ by Charpit's Method

Soln: Given, $z - px - qy = p^2 + q^2$ — ①

$$\therefore f \equiv z - px - qy - p^2 - q^2 = 0$$

$$\frac{\partial f}{\partial x} = -p - q$$

$$\frac{\partial f}{\partial y} = -q$$

$$\frac{\partial f}{\partial z} = 1$$

$$\frac{\partial f}{\partial p} = -x - 2p$$

$$\frac{\partial f}{\partial q} = -y - 2q$$

Charpit's auxiliary equations are,

$$\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \dots$$

$$\Rightarrow \frac{dp}{-p+p} = \frac{dq}{-q+q} = \dots$$

$$\Rightarrow \frac{dp}{0} = \frac{dq}{0} = \dots$$

Thus from 1st $dp = 0$

Integrating we get.

$p = a$ where a is an arbitrary constant.

Similarly from 2nd we get,
 $q = b$ where b is an arbitrary constant.

from (1) $z - ax - by = a^2 + b^2$
 $\therefore z = ax + by + a^2 + b^2$

[Answer]

Ex Solve $z = px + qy + pq$

Soln: Given, $z = px + qy + pq$ — (1)

$\therefore f \equiv z - px - qy - pq = 0$

$\therefore \frac{\partial f}{\partial x} = -p$

$\frac{\partial f}{\partial y} = -q$

$\frac{\partial f}{\partial z} = 1$

$\frac{\partial f}{\partial p} = -x - q$

$\frac{\partial f}{\partial q} = -y - p$

Charpit's auxiliary equations are,

$$\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{dF}{0}$$

$\Rightarrow \frac{dp}{-p+p} = \frac{dq}{-q+q} = \dots$

$\Rightarrow \frac{dp}{0} = \frac{dq}{0} = \dots$

Thus from 1st

$dp = 0$

Integrating, we get

$p = a$ where a is an arbitrary constant

Similarly from 2nd we get

$q = b$ where b is an arbitrary constant.

from (1)

$z = px + qy + pq$

$\therefore z = ax + by + ab$

[Answer]

Q Solve $P(1+q^2) = q(z-a)$

Given, $P(1+q^2) = q(z-a)$ — (1)

$\therefore f = P + pq^2 - qz + qa = 0$

$\frac{\partial f}{\partial z} = 0$

$\frac{\partial f}{\partial y} = 0$

$\frac{\partial f}{\partial z} = -q$

$\frac{\partial f}{\partial p} = 1 + q^2$

$\frac{\partial f}{\partial q} = 2pq - z + a$

Charpit's auxiliary equations are,

$$\frac{dp}{\frac{\partial f}{\partial z} + p \frac{\partial f}{\partial p}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial p}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{df}{0}$$

$$\therefore \frac{dp}{-pq} = \frac{dq}{-q^2} = \frac{dz}{3pq^2 + p(a-z)q} = \frac{dz}{q^2 + 1} = \frac{dy}{-z + a + 2pq} = \frac{df}{0}$$

From 1st two

$$\frac{dp}{p} = \frac{dq}{q}$$

Integrating, we get $\ln p = \ln q + \ln c$ where c is an arbitrary constant

$\Rightarrow \ln p = \ln c q$

$\Rightarrow p = cq$ — (2)

Putting $p = cq$ into (1)

$cq(1+q^2) = q(z-a)$

$\Rightarrow cq + cq^3 - qz + qa = 0 \Rightarrow c + cq^2 = z - a \Rightarrow cq^2 = z - a - c$

$\Rightarrow q = \sqrt{c(z-a)-1} \sqrt{\frac{z-a-c}{c}}$

From (2) $p = c \cdot \sqrt{c(z-a)-1} \sqrt{\frac{z-a-c}{c}}$

$= \sqrt{c} \cdot \sqrt{z-a-c}$

From $dz = p dx + q dy$

$\Rightarrow dz = \sqrt{c} \cdot \sqrt{z-a-c} dz + \frac{\sqrt{z-a-c}}{\sqrt{c}} dy$

$\Rightarrow \frac{dz}{\sqrt{z-a-c}} = \sqrt{c} dz + \frac{1}{\sqrt{c}} dy$

$\Rightarrow \frac{dm}{\sqrt{m}} = \sqrt{c} dx + \frac{1}{\sqrt{c}} dy$

Let $\begin{cases} z-a-c = m \\ dz = dm \end{cases}$

Integrating we get $\frac{2}{3} m^{3/2} = \sqrt{c} x + \frac{1}{\sqrt{c}} y + d$ where d is an arbitrary constant.

$\therefore \frac{2}{3} (z-a-c)^{3/2} = \sqrt{c} x + \frac{1}{\sqrt{c}} y + d$

Answer

12 September, 2022

Solve $z = pq$ by Charpit's method.

Sol: Given, $z = pq$ — (i)
 $\therefore f = z - pq = 0$

$$\frac{\partial f}{\partial z} = 0$$

$$\frac{\partial f}{\partial p} = -q$$

$$\frac{\partial f}{\partial q} = -p$$

$$\frac{\partial f}{\partial p} = -q$$

$$\frac{\partial f}{\partial q} = -p$$

Charpit's auxiliary equations are,

$$\begin{aligned} \frac{dp}{\frac{\partial f}{\partial p}} &= \frac{dq}{\frac{\partial f}{\partial q}} = \frac{dz}{-\frac{\partial f}{\partial z}} = \frac{dx}{\frac{\partial f}{\partial x}} = \frac{dy}{\frac{\partial f}{\partial y}} = \frac{df}{0} \\ &= \frac{dp}{0-p} = \frac{dq}{0-q} = \frac{dz}{-pq-pq} = \frac{dx}{q} = \frac{dy}{p} = \frac{df}{0} \end{aligned}$$

From 1st two

$$\frac{dp}{p} = \frac{dq}{q}$$

Integrating we get,

$$\ln p = \ln q + \ln c \quad \text{where } c \text{ is an arbitrary constant}$$

$$\ln p = \ln cq$$

$$\Rightarrow p = cq \text{ — (ii)}$$

Putting $p = cq$ in (i)

$$z = cq \cdot q$$

$$\Rightarrow z = cq^2$$

$$\Rightarrow q^2 = \frac{z}{c}$$

$$\Rightarrow q = \sqrt{\frac{z}{c}}$$

from (ii)

$$p = c \cdot \sqrt{\frac{z}{c}}$$

$$= \sqrt{c} \cdot \sqrt{z}$$

$$= \sqrt{cz}$$

from, $dz = p dx + q dy$

$$= \sqrt{cz} dx + \sqrt{\frac{z}{c}} dy$$

Ex Solve $z = (xy+z)^2$ by Charpit's method.

Given, $p = (xy+z)^2$

$$= q^2xy + 2qyz + z^2$$

$$\Rightarrow f \equiv p - q^2xy - 2qyz - z^2$$

$$\frac{\partial f}{\partial z} = 0$$

$$\frac{\partial f}{\partial y} = -2xyq - 2qz$$

$$\frac{\partial f}{\partial x} = -2xy - 2z$$

$$\frac{\partial f}{\partial p} = 1$$

$$\frac{\partial f}{\partial q} = -2xyq - 2yz$$

Charpit's auxiliary equations are,

$$\frac{dp}{\frac{\partial f}{\partial p}} = \frac{dq}{\frac{\partial f}{\partial q}} = \frac{dz}{\frac{\partial f}{\partial z}} = \frac{dx}{\frac{\partial f}{\partial x}} = \frac{dy}{\frac{\partial f}{\partial y}}$$

from 1st and fifth,

$$\frac{dp}{-2q(xy+z)} = \frac{dy}{2y(xy+z)}$$

$$\Rightarrow \frac{dp}{-p} = \frac{dy}{y}$$

$$\Rightarrow \frac{dp}{p} + \frac{dy}{y} = 0$$

Integrating, we get,

$$\ln p + \ln y = \ln c$$

$$\Rightarrow py = c$$

$$\Rightarrow p = \frac{c}{y}$$

Putting $p = \frac{c}{y}$ into given equation.

$$\frac{c}{y} = (xy+z)^2$$

$$\Rightarrow xy+z = \sqrt{\frac{c}{y}}$$

$$\Rightarrow xy = \sqrt{\frac{c}{y}} - z$$

$$\therefore x = \frac{1}{y} (\sqrt{\frac{c}{y}} - z)$$

Now from, $dz = p dx + q dy$

$$\Rightarrow dz = \frac{c}{y} dx + \frac{1}{y} (\sqrt{\frac{c}{y}} - z) dy$$

$$\Rightarrow y dz = c dx + \sqrt{\frac{c}{y}} dy - z dy$$

$$\Rightarrow y dz + z dy = c dx + \sqrt{\frac{c}{y}} dy$$

$$\Rightarrow d(yz) = c dx + \sqrt{\frac{c}{y}} \cdot \frac{dy}{y}$$

Integrating, we get,

$$\int d(yz) = c \int dx + \sqrt{c} \int \frac{dy}{\sqrt{y}}$$

$$\Rightarrow yz = cx + 2\sqrt{y} \cdot \sqrt{c} + m$$

$$\Rightarrow yz = cx + 2\sqrt{y}c + m \text{ --- where } c \text{ and } m \text{ are arbitrary constant.}$$

22 September, 2022

Solve, $x^2p + y^2q = \frac{z^2}{xy}$ by Lagrange method.

Given, $x^2p + y^2q = \frac{z^2}{xy}$

Lagrange A.E.

$$\frac{dx}{x^2} = \frac{dy}{y^2} = \frac{dz}{x^2y^2}$$

Lagrange Auxiliary Equation

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$* \phi(u, v) = 0$$

From 1st two,

$$\frac{dx}{x^2} = \frac{dy}{y^2}$$

$$\Rightarrow \frac{dx}{x} = \frac{dy}{y}$$

Integrating we get,

$$\ln x = \ln y + \ln c_1$$

$$\Rightarrow x = y c_1$$

$$\therefore c_1 = \frac{x}{y} \text{ --- (1)}$$

From last two,

$$\frac{dy}{y^2} = \frac{dz}{z^2}$$

$$\Rightarrow \frac{dy}{y} = \frac{dz}{z}$$

Integrating we get,

$$\ln y = \ln z + \ln c_2$$

$$\Rightarrow \ln y = \ln z + \ln c_2$$

$$\Rightarrow y = c_2 z$$

$$\therefore c_2 = \frac{y}{z} \text{ --- (2)}$$

where c_1, c_2 are two arbitrary constants.

Thus complete integral on solution.

$$\phi(c_1, c_2) = 0$$

$$\Rightarrow \phi\left(\frac{x}{y}, \frac{y}{z}\right) = 0$$

[Answer]

Ex Solve. $(z^2 - yz)p + (y^2 - zx)q = z^2 - xy$ by Lagrange method

Given,

Lagrange A.E.

$$\frac{dx}{x^2 - yz} = \frac{dy}{y^2 - zx} = \frac{dz}{z^2 - xy}$$

$$\Rightarrow \frac{dz - dy}{x^2 - yz} = \frac{dy - dz}{y^2 - zx} = \frac{dx - dz}{z^2 - xy}$$

$$\Rightarrow \frac{dx}{x^2 - yz} = \frac{dy}{y^2 - zx} = \frac{dz}{z^2 - xy} = \frac{dx - dy}{(x+y)(x-y)z(x-y)} = \frac{dy - dz}{(y+z)(y-z)+x(y-z)}$$

$$= \frac{dx - dz}{(x+z)(x-z)y(x-z)}$$

$$= \frac{dx}{x^2 - yz} = \frac{dy}{y^2 - zx} = \frac{dz}{z^2 - xy} = \frac{dx - dy}{(x-y)(x+y+z)} = \frac{dy - dz}{(y-z)(x+y+z)}$$

$$= \frac{dx - dz}{(x-z)(x+y+z)}$$

From 4th and 5th

$$\frac{dx - dy}{(x-y)(x+y+z)} = \frac{dy - dz}{(y-z)(x+y+z)}$$

$$\Rightarrow \frac{dx - dy}{x-y} = \frac{dy - dz}{y-z}$$

Similar Integrating we get,

$$\ln(x-y) = \ln(y-z) + \ln c_1$$

$$\Rightarrow \ln(x-y) = \ln c_1(y-z)$$

$$\therefore c_1 = \frac{x-y}{y-z}$$

Similarly from 5th and 6th,

$$c_2 = \frac{y-z}{x-z}$$

Thus complete integral is,

$$\phi(c_1, c_2) = 0$$

$$\Rightarrow \phi\left(\frac{x-y}{y-z}, \frac{y-z}{x-z}\right) = 0$$

Q1 Solve, $(mz - ny)p + (nx - lz)q = y - mx$ by Lagrange method.

Given that,

$$(mz - ny)p + (nx - lz)q = y - mx.$$

Lagrange A.E.

$$\frac{dx}{mz - ny} = \frac{dy}{mx - lz} = \frac{dz}{y - mx} = \frac{ldx + mdy + ndz}{0} = \frac{x dx + y dy + z dz}{0}$$

From 4th, $ldx + mdy + ndz = 0$

Integrating we get,
 $lx + my + nz = c_1$

From 5th, $x dx + y dy + z dz = 0$

Integrating we get,
 $\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = \frac{c_2}{2}$
 $\therefore x^2 + y^2 + z^2 = c_2$

where, c_1 and c_2 are arbitrary constant.

Thus complete Integral solution is,

$$\Phi(c_1, c_2) = 0$$

$$\Phi(lx + my + nz, x^2 + y^2 + z^2) = 0$$

[Answer]

26 September 2020

Q2 Solve $(z^2 - 2yz - y^2)p + (xy + xz)q = xz - x^2$ by Lagrange method.

Solution: Given that,

$$(z^2 - 2yz - y^2)p + (xy + xz)q = xz - x^2$$

Lagrange Auxiliary equation,

$$\frac{dx}{z^2 - 2yz - y^2} = \frac{dy}{xy + xz} = \frac{dz}{xz - x^2} = \frac{xdx}{xz^2 - x^2z - y^2x} = \frac{xdx}{x(z^2 - 2yz - y^2)}$$

$$= \frac{dz}{xz^2 - x^2z} = \frac{xdx + ydy + zdz}{xz^2 - 2xyz - x^2z} = \frac{xdx + ydy + zdz}{x(z^2 - 2yz - y^2)}$$

$$\Rightarrow \frac{dx}{z^2 - 2yz - y^2} = \frac{dy}{x(y + z)} = \frac{dz}{x(z - x)} = \frac{xdx}{x^2(z - x)(y + z)}$$

$$= \frac{xdx}{x^2(z - x)(y + z)} = \frac{dz}{x(z - x)(y + z)} = \frac{xdx + ydy + zdz}{x^2(z - x)(y + z)}$$

From 2nd and 3rd,

$$\frac{dy}{x(y + z)} = \frac{dz}{x(z - x)}$$

$$\Rightarrow ydy - zdy = ydz + zdz$$

$$\Rightarrow ydy - zdy - ydz - zdz = 0$$

$$\Rightarrow ydy - zdy - d(yz) = 0$$

Integration we get,

$$\frac{y^2}{2} - \frac{z^2}{2} - yz = \frac{C_1}{2}$$

$$\Rightarrow y^2 - z^2 - 2yz = C_1$$

From 4th,

$$x dx + y dy + z dz = 0$$

$$\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = \frac{C_2}{2}$$

$$\therefore x^2 + y^2 + z^2 = C_2$$

Hence complete solution is, $\phi(C_1, C_2) = 0$

$$\therefore \phi(y^2 - z^2 - 2yz, x^2 + y^2 + z^2) = 0$$

Q Solve $x(y^2+z)P - y(x^2+z)Q = z(x^2-y^2)$ by L.M.

Solution: Given that

$$x(y^2+z)P - y(x^2+z)Q = z(x^2-y^2)$$

Lagrange Auxiliary equation,

$$\frac{dx}{x(y^2+z)} = \frac{dy}{-y(x^2+z)} = \frac{dz}{z(x^2-y^2)}$$

$$\Rightarrow \frac{dx}{x y^2 + x z^2} = \frac{dy}{-x^2 y - y z^2} = \frac{dz}{x^2 z - y^2 z} = \frac{x dx + y dy - dz}{x^2 z - x^2 y - x^2 z + x^2 y - y^2 z + y^2 z}$$

$$= \frac{\frac{dx}{x} + \frac{dy}{y} - \frac{dz}{z}}{y^2 - x^2 - z^2 + x^2 - y^2}$$

$$\Rightarrow \frac{dx}{x y^2 - x z^2} = \frac{dy}{-x^2 y - y z^2} = \frac{dz}{x^2 z - y^2 z} = \frac{x dx + y dy - dz}{0} = \frac{\frac{dx}{x} + \frac{dy}{y} - \frac{dz}{z}}{0}$$

from 4th,

$$x dx + y dy - dz = 0$$

Integrating,

$$\frac{x^2}{2} + \frac{y^2}{2} - z = \frac{C_1}{2}$$

$$\Rightarrow x^2 + y^2 - 2z = C_1$$

From 5th,

$$\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = 0$$

Integrating, $\ln x + \ln y + \ln z = \ln C_2$

$$xyz = C_2$$

Hence a complete solution is

$$\phi(C_1, C_2) = 0$$

$$\therefore \phi(x^2 + y^2 - 2z, xyz) = 0$$

Q1 Solve $P \cos(x+y) + Q \sin(x+y) = Z$

Soln: Given that,

$$P \cos(x+y) + Q \sin(x+y) = 0$$

Lagrange's auxiliary equation are,

$$\frac{dx}{\cos(x+y)} = \frac{dy}{\sin(x+y)} = \frac{dz}{Z} \quad \text{--- (i)}$$

Now to find two integrals of (i) we have

$$\frac{dx+dy}{\cos(x+y) + \sin(x+y)} = \frac{dx-dy}{\cos(x+y) - \sin(x+y)}$$

$$\text{or, } \frac{[-\sin(x+y) + \cos(x+y)](dx+dy)}{\cos(x+y) + \sin(x+y)} = dx-dy$$

Now numerator on the left is differential of the denominator:

$$\text{Integrating } \log [\cos(x+y) + \sin(x+y)] = x-y + \log a$$

$$\text{or, } [\cos(x+y) + \sin(x+y)] e^{y-x} = a \quad \text{--- (ii)}$$

$$\text{again, } \frac{dx+dy}{\cos(x+y) + \sin(x+y)} = \frac{dz}{Z}$$

$$\frac{dx+dy}{\sqrt{2} \sin(x+y + \frac{1}{4}\pi)} = \frac{dz}{Z}$$

$$\text{Integrating, } \frac{1}{2} \log \tan \left[\frac{1}{2}(x+y) + \frac{1}{8}\pi \right] = \log z + \log b$$

$$\therefore \tan \left[\frac{1}{2}(x+y) + \frac{1}{8}\pi \right] z^{-\sqrt{2}} = b^{\sqrt{2}} \quad \text{--- (iii)}$$

Hence the solution of given equation is,

$$\phi \left[\{ \cos(x+y) + \sin(x+y) \} e^{y-x}, z^{-\sqrt{2}} \tan \left\{ \frac{1}{2}(x+y) + \frac{1}{8}\pi \right\} \right] = 0$$